Optimizing Chunk Size and Overlap in a FAISS-Based Retrieval-Augmented Generation System for Philippine Labor-Benefit Question Answering

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***Abstract*—This study explores the use of retrieval-augmented generation (RAG) for answering questions based on the *Workers' Statutory Monetary Benefits Handbook, 2024 Edition*. The system was designed to help retrieve relevant labor-benefit provisions and generate concise answers grounded in the handbook. Document passages were divided into overlapping chunks, converted into vector embeddings, and stored in a FAISS index. Candidate passages were retrieved and reranked before the three most relevant chunks were provided to Qwen3-1.7B for answer generation. The experiment was conducted in Google Colab using a T4 GPU. Three questions representing easy, average, and difficult levels were evaluated by comparing the generated answers with manually verified provisions from the handbook. The system answered the easy and average questions correctly, including the applicable pay percentages and source pages. For the difficult question, the correct passages were retrieved, but the model was unable to combine the premium-pay and overtime-pay rules accurately. This observation shows that retrieving relevant evidence does not always guarantee a correct answer, particularly when several related rules and calculations must be considered together. Although limited to three questions, the activity provides a practical view of both the usefulness and limitations of a small-model RAG system for document-based labor-benefit inquiries.**

Keywords*—retrieval-augmented generation, FAISS, Qwen3-1.7B , document retrieval, labor benefits, question answering.

I. Introduction

Understanding statutory monetary benefits can be challenging because the correct answer often depends on several details about a worker's situation. A regular holiday, a special non-working day, a scheduled rest day, and overtime work may each have different compensation rules. Although these rules are explained in the *Workers' Statutory Monetary Benefits Handbook, 2024 Edition*, finding the appropriate provision can still require careful reading, especially when similar pay rates appear on neighboring pages.

This challenge encouraged the exploration of retrieval-augmented generation as a possible approach to document-based question answering. Retrieval-augmented generation, commonly referred to as RAG, allows a language model to answer a question using passages retrieved from an external document instead of relying only on information learned during model training. This is particularly useful for labor-benefit questions because an answer should remain connected to the handbook rather than depend on unsupported assumptions. In this setting, retrieving the correct passage matters just as much as producing a clear response.

This study developed a RAG system that uses FAISS for vector retrieval and Qwen3-1.7B for answer generation. The system was tested using the 2024 handbook as its knowledge source. Three questions were prepared at easy, average, and difficult levels to observe how the system handled a directly stated rule, a rule with several closely related conditions, and a question requiring information from multiple passages.

The study aimed to determine whether the system could retrieve the appropriate handbook evidence, generate an answer that accurately reflected that evidence, and provide the correct source page. It also examined what happened when the model needed to distinguish between similar provisions and combine multiple pay formulas. Rather than treating the experiment as a broad measurement of accuracy, the study used the three questions as a small exploratory test of where the system worked reliably and where further improvement was still needed.

II. RAG and Model

A. Retrieval-Augmented Generation

Retrieval-augmented generation combines information retrieval with language-model generation. When a question is submitted, the system first searches an indexed document collection for relevant passages. The selected passages are then included in the prompt given to the language model.

This allows the generated answer to remain connected to the source document instead of depending entirely on information learned during model training.

This approach was suitable for the activity because the answers needed to come directly from the labor-benefits handbook. It was not enough for the model to produce a reasonable-sounding response. The applicable conditions, percentages, formulas, and page references also needed to agree with the document.

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B. Text Embeddings and FAISS

Text embeddings represent passages and questions as numerical vectors based on their meaning. Passages with meanings that are similar to the question are expected to appear closer to the question within the vector space. In this study, the sentence-transformers/all-MiniLM-L6-v2 model was used to generate embeddings for the handbook chunks and user questions.

FAISS was used to store and search the document vectors. For every submitted question, it compared the question vector with the stored chunk vectors and returned the most similar candidates. This made it possible to search the handbook according to meaning instead of depending only on exact keyword matches.

C. Cross-Encoder Reranking

Semantic similarity alone may retrieve passages that contain related words but describe a different situation. For example, passages about regular holidays, special non-working days, rest days, and overtime may use similar terms while applying different pay rates. To improve the ordering of the retrieved passages, the system used the cross-encoder/ms-marco-MiniLM-L-6-v2 reranker.

The reranker evaluated the complete question together with each candidate passage. It then assigned a relevance score and reordered the candidates. The three highest-ranked chunks were included as evidence in the final prompt. This additional step was important because the quality of the generated answer depended heavily on whether the most applicable passage appeared among the selected evidence.

D. Qwen3-1.7B

Qwen3-1.7B was used as the answer-generation model. Its relatively compact size allowed it to run using the T4 GPU available through the free tier of Google Colab. The model received the original question, the three highest-ranked handbook passages, and instructions to answer only from the supplied evidence.

The model was also instructed to distinguish between an additional premium and a total pay rate, avoid combining unrelated labor rules, and admit when the retrieved evidence did not directly support an answer. These instructions were included because labor-benefit provisions can appear very similar even when they apply to different working conditions. The experiment therefore examined not only whether Qwen could generate a readable response, but also whether it could interpret the retrieved rules carefully and apply them to the correct situation.

III. Methodology

A. Model Selection

The initially proposed generation model for the activity was gpt-oss-20b. However, the GPU resources available through the free tier of Google Colab were not sufficient for running the model under the intended setup. Qwen3-1.7B was therefore selected as a smaller alternative that could operate using the available T4 GPU. This adjustment allowed the experiment to continue while keeping the main purpose of the activity unchanged.

B. Source Document

The knowledge source used in the experiment was the *Workers' Statutory Monetary Benefits Handbook, 2024 Edition*. The handbook contains information about minimum wage, holiday pay, premium pay, overtime pay, leave benefits, and other forms of statutory compensation. It was selected because its provisions contain several conditions and formulas that are useful for evaluating document retrieval and evidence-based answer generation.

The PDF page number of each extracted passage was retained during processing. This allowed the generated answers to be checked against the original handbook and made it easier to determine whether the cited page actually supported the response.

C. Document Preprocessing

The handbook text was extracted from the uploaded PDF and divided into smaller passages before indexing. The chunk size was set to 1,000, with an overlap of 200 between consecutive chunks. The overlap was used to reduce the possibility of losing information when a provision continued across a chunk boundary. Each chunk was assigned a unique identifier and linked to its PDF page number. The processed chunks were then prepared for embedding and storage in the vector index.

D. Embedding and Vector Indexing

The sentence-transformers/all-MiniLM-L6-v2 model converted each text chunk into a normalized numerical vector. The vectors were stored in a FAISS index together with the corresponding chunk identifiers, page numbers, and original text. When a question was entered, the same embedding model converted it into a query vector. FAISS compared the query vector with the stored document vectors and returned the 30 most similar candidate chunks. Using the same embedding model for both the document and the question kept their vector representations comparable.

E. Retrieval and Reranking

The 30 FAISS candidates were passed to the cross-encoder/ms-marco-MiniLM-L-6-v2 reranker. Unlike the embedding search, which compared stored vectors, the cross-encoder examined the complete question and passage together. It assigned a relevance score to every

candidate and reordered them according to their relationship with the question.

The three highest-ranked chunks were selected as the final evidence. Their page numbers, chunk identifiers, FAISS ranks, and reranker scores were displayed in the notebook to make the retrieval process easier to inspect.

F. Answer Generation

The selected evidence and original question were inserted into a structured prompt and submitted to Qwen3-1.7B. The model was instructed to act as a document-grounded Philippine labor-benefits information assistant and to answer only from the supplied handbook evidence.

The instructions required the model to identify the applicable labor benefit, distinguish an additional premium from a total pay rate, and avoid combining rules for different working conditions. When the evidence did not directly support an answer, the model was instructed to state that the answer could not be found in the provided document. The application then added the PDF page number associated with the highest-ranked source.

G. Evaluation Procedure

Three English questions were prepared at easy, average, and difficult levels. The easy question required one directly stated holiday-pay rule. The average question required the system to distinguish a special non-working day falling on a rest day from similar premium-pay conditions. The difficult question required the use of both premium-pay and overtime-pay provisions and included a multi-step calculation.

For every question, the expected answer was first verified manually using the handbook. The generated answer was then compared with the expected percentage, formula, applicable conditions, and supporting page. The retrieved passages were also inspected to determine whether an incorrect response resulted from retrieval failure or from the language model's interpretation of otherwise correct evidence..

IV. Experimental Configuration

The experiment was conducted in Google Colab using the T4 GPU available through its free tier. The same notebook configuration was maintained for the easy, average, and difficult questions so that the observed differences could be related to question complexity rather than changes in the retrieval settings.

TABLE I. Experiment Configurations

Component	***Configuration***
Source document	Workers' Statutory Monetary Benefits Handbook, 2024 Edition
Generation model	Qwen/Qwen3-1.7B
Embedding model	sentence-transformers/all-MiniLM-L6-v2
Vector store	FAISS
Reranking model	cross-encoder/ms-marco-MiniLM-L-6-v2
Chunk size	1,000
Chunk overlap	200
FAISS candidate count	30
Final evidence count	Top 3 reranked chunks
Question language	English
Execution environment	Google Colab free tier
Hardware accelerator	NVIDIA T4 GPU

The notebook also displayed retrieval and generation diagnostics for every question. These included the FAISS candidate count, reranking scores, selected chunk identifiers, source pages, prompt length, and the evidence supplied to Qwen3-1.7B. Keeping these details visible was useful when an answer was incorrect because it helped show whether the relevant passage had not been retrieved or whether the model had misinterpreted evidence that was already available.

V. Experimental Configuration

TABLE II. Experiment Configurations

Difficulty	***Expected Answer***	***Generated Answer***	***Retrieved Evidence***	***Result***
Easy	Total pay of 200% of the basic wage	Returned a total pay rate of 200%	PDF page 28 ranked first	Correct
Average	Additional 50%; total pay of 150%	Returned an additional 50% and total pay of 150%	PDF page 32 ranked first	Correct
Difficult	First eight hours at 260%; two overtime hours equivalent to 84.5% of the basic daily wage; total pay of 344.5%	Returned 150% for the first eight hours and an incorrect total of 200%	Relevant rules from PDF pages 32 and 33 were retrieved	Incorrect

The easy question asked for the total pay rate for the first eight hours of work on a regular holiday. The system ranked PDF page 28 first and correctly returned 200% of the basic wage. For the average question, the system identified PDF page 32 as the highest-ranked source and correctly stated that work performed during the first eight hours of a special non-working day falling on an employee's scheduled rest day receives an additional 50%, resulting in total pay of 150%. The difficult question required the system to combine the first-eight-hours rule from PDF page 32 with the overtime rule from PDF page 33. Although both relevant pages were retrieved, the model

incorrectly returned 150% for the first eight hours and a total of 200%. The expected answer was 260% for the first eight hours, 84.5% of the basic daily wage for the two overtime hours, and a combined total of 344.5%. Overall, the system answered two of the three questions correctly, representing 66.7% of the small evaluation set. This percentage reflects only the recorded activity and should not be interpreted as a general measure of the system's accuracy.

References

[1] DOLE Bureau of Working Conditions, "Handbook on Workers' Statutory Monetary Benefits," 2024 Edition. Available: bwc.dole.gov.ph